

Large language models as linguistic simulators and cognitive models in human research

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Abstract

The rise of large language models (LLMs) that generate human-like text has sparked debates over their potential to replace human participants in behavioral and cognitive research. We critically evaluate this replacement perspective to appraise the fundamental utility of language models in psychology and social science. Through a five-dimension framework—characterization, representation, interpretation, implication, and utility—we identify six fallacies that undermine the replacement perspective: (1) equating token prediction with human intelligence, (2) assuming LLMs represent the average human, (3) interpreting alignment as explanation, (4) anthropomorphizing AI, (5) essentializing identities, and (6) purporting LLMs as primary tools that directly reveal the human mind. Rather than replacement, the evidence and arguments are consistent with a simulation perspective, where LLMs offer a new paradigm to simulate roles and model cognitive processes. We highlight limitations and considerations about internal, external, construct, and statistical validity, providing methodological guidelines for effective integration of LLMs into psychological research—with a focus on model selection, prompt design, interpretation, and ethical considerations. This perspective reframes the role of language models in behavioral and cognitive science, serving as linguistic simulators and cognitive models that shed light on the similarities and differences between machine intelligence and human cognition and thoughts.

Keywords: Generative AI (GenAI), large language models (LLMs), AI participants (AI subjects), silicon sampling, simulation or modeling (modelling), latent psychology and latent cognition

Language models as human participants

Trained with an extensive corpus of online text, large language models (LLMs) output responses that are ostensibly human-like (**Box 1**). In many perceptual, linguistic, cognitive, and moral reasoning tasks, LLMs generate responses that closely capture what average people perceive, say, think, or do ¹⁻⁴. This unique combination—extensive knowledge and human-like interaction—has led to an intriguing proposal: using LLMs to replace human participants in behavioral research. Across various fields, LLMs are heralded to “substitute human participants” in empirical research as “silicon samples” ⁵; “supplant human participants for data collection” in social science as “simulated participants” ⁶; “replace human participants” in psychological science as “synthetic AI participants” ¹; and “replace humans” in human-centered design to provide “simulated user responses” ⁷. Echoing this perspective, a company offers human-like AI participants for user and market research—advertising user research “without the users.”

The idea of using LLMs to replace human participants reflects a broader narrative in AI, one that aims to replace humans with artificial counterparts by creating autonomous AI systems that “outperform humans at most economically valuable work” ⁸ or “on a vast array of tasks” ⁹. It is therefore of particular importance to thoroughly evaluate the replacement perspective and its underlying assumptions, and to properly situate LLMs in the landscape of behavioral and social science research. We synthesize five key dimensions that assess the scientific and philosophical validity of applying LLMs in social and behavioral studies: characterization (the nature of intelligence in LLMs), representation (how accurately LLMs capture human thoughts, attitudes, and behaviors), interpretation (what alignment between humans and LLMs tells us about their mechanisms), implication (the broader ethical and societal consequences), and utility (the practical limitations and benefits of using LLMs in human research).

Using this framework, we identify six fallacies that misinterpret LLMs under the replacement perspective, demonstrating its fundamental untenability: the token prediction as human intelligence fallacy; the average human fallacy; the alignment as explanation fallacy; the anthropomorphism fallacy; the identity essentialization fallacy; and the substitution fallacy. The synthesis supports a simulation perspective, where LLMs complement but do not—and cannot—replace human participants, functioning as neural language simulators that help reveal embedded psychological and cognitive dynamics (**Table 1**). Such dynamics—latent psychology and cognition—are rooted in textual communication that embodies human thoughts, attitudes, and behaviors ^{10,11}.

To harness the potential of LLMs in human research, we assess their validity across four dimensions: internal validity, external validity, construct validity, and statistical conclusion validity ¹². This analysis highlights the methodological rigor required for their integration into empirical research (**Table 2**). Based on these insights, we offer guidelines for leveraging LLMs as linguistic simulators and cognitive models (**Table 3**).

Box 1 | Language models and the power of language

AI chatbots such as ChatGPT are constructed on the foundation of LLMs, which have seen major advancements since the introduction of the transformer architecture ¹³. The architecture’s self-attention mechanism allows LLMs to weigh the importance of each word in context, improving comprehension and generation capabilities. A specialized attention mechanism—the “induction heads”—enables LLMs to generalize patterns by matching current tokens to prior similar tokens in a sequence and predicting the next

token. By representing tokens as context-dependent vectors in a high-dimensional space, LLMs capture complex relationships between words, phrases, and concepts, allowing them to go beyond traditional distributional semantics¹⁴ to model the nuanced, contextualized meanings and structures in language¹⁵⁻¹⁷, including visual knowledge^{18,19} and concepts like space and time²⁰.

To more closely align with human values and preferences (e.g., being accurate, pro-social, and helpful), base models (also known as foundation models or pretrained models) are often further fine-tuned using techniques like reinforcement learning from human feedback (RLHF)²¹. While base models typically employ self-supervised learning on massive text datasets to model language (e.g., predict the next token in GPT; predict masked tokens or next sentences in BERT), fine-tuning adapts the base model for a specific downstream task or domain using various methods. Instruct models like ChatGPT, for example, use both supervised learning with a smaller, task-specific dataset (e.g., text classification or instruction–response pairs) and reinforcement learning (e.g., training a dialogue agent to be more engaging).

The ability of LLMs to generate natural, context-aware responses through mechanisms such as RLHF contributes to interactions that often feel deeply human. When a chatbot writes a story that moves us to tears or replies with empathy, it is only natural to project our human experiences onto it—that it understands pain, love, and fear, that it cares about us and is happy to talk to us. In a 1923 speech to the Royal College of Surgeons in London, Rudyard Kipling observed, “*Words are, of course, the most powerful drug used by mankind. Not only do words infect, egotize, narcotize, and paralyze, but they enter into and color the minutest cells of the brain.*” While Kipling did not envision the debut of LLMs, the words generated by AI systems can be just as seductive—the smooth, natural conversations evoke an immersive experience of chatting with another human. But this would be a mistake: to confuse a language model with the human users of language.

Table 1 | Conceptualizing language models in social and behavioral sciences as participant replacements versus simulators

Dimension	LLMs as replacement for human participants	LLMs as linguistic simulators of human responses	Core argument	Potential fallacy
Characterization	A singular machine mind that encodes human judgments	A multifaceted tool that simulates diverse perspectives	LLMs optimize next-token prediction, producing machine intelligence distinct from biological intelligence	Token prediction as human intelligence fallacy
Representation	LLMs are assumed to approximate humans,	LLMs’ behavior depends on training and	LLMs are engineered to surpass average human	The average human fallacy

	especially Western English speakers	fine-tuning, not assumption of average responses or human-like cognition	performance, trained on vast but biased text, and fine-tuned for accuracy and helpfulness	
Interpretation	Correlational alignment between humans and LLMs means shared mechanisms	Correlation does not mean mechanistic equivalence	LLMs model aspects of human cognition, but differ fundamentally in training and architecture	Alignment as explanation fallacy
Implication	Lead to AI anthropomorphism and identity essentialization	Clarify LLMs' nature as statistical models capable of role-playing and cognitive simulation	Anthropomorphism can create unfounded fears and unrealistic expectations about LLMs	Anthropomorphism fallacy and identity essentialization fallacy
Utility	LLMs can be primary tools for directly revealing the human mind	LLMs are supplementary tools with key limitations due to training data, algorithms, fine-tuning, and closed models	LLMs' responses must be corroborated with actual human data to validate insights	Substitution fallacy

Six fallacies that misinterpret language models

In analyzing theoretical arguments and empirical evidence bearing on the replacement perspective, we found that beneath its economic and sci-fi attractiveness lie six major fallacies.

Token prediction as human intelligence fallacy

By compressing vast amounts of text data into network weights, LLMs predict the next tokens in a sequence, determining which words are most likely to follow a prompt based on the distribution of the training data and fine-tuning data. Remarkably, optimization for next-token prediction—combined with the ability to handle abstract, context-based learning—enables tasks once thought to exclusively require human intelligence, including some even unanticipated by their developers. In particular, this process equips LLMs with intricate

linguistic knowledge, including semantic relations and syntactic rules—namely, *formal* linguistic competence¹⁶. It allows LLMs to transcend mere memorization and tackle complex, context-dependent tasks in language processing and generation^{16,22}. It also enables LLMs to infer underlying task structures and generate contextually appropriate responses—namely, *instrumental* knowledge²³.

Yet, at their core, LLMs have no minds²⁴ but are statistical models that optimize next-token prediction—a task that is fundamentally different from those of humans, thus producing a kind of singular, ungrounded intelligence that fundamentally differs from biological intelligence. Human intelligence integrates general and specialized capabilities and is a product of evolutionary adaptations and developmental learning (**Box 2**). It is “grounded in one’s embodied physical and emotional experiences” and “deeply reliant on one’s social and cultural environments”²⁵. In ways that symbols in symbolic systems lack real-world referents, tokens in LLMs also lack real-world referents, meaning, and experience²⁶. This lack of grounding detaches the models from the very physical and social realities they simulate, hindering them from obtaining *functional* linguistic competence—the use of language to achieve goals in the world¹⁶—and from acquiring the *worldly* knowledge necessary to approximate world models—representations of the real world that are structure-preserving and behaviorally efficacious^{23,27}, such as cognitive maps, body schemas, or spatial schemas²⁸.

Thus, while LLMs manifest proficient language *use* that contrasts with simply searching pre-recorded strings of text (a lookup table of all possible conversations, as in Block’s hypothetical Blockhead machine), they are not true language *users*: they do not possess intrinsic meaning, communicative intentions, or other internal states essential to human language users²⁹. By generating coherent language without the kind of intentionality, consciousness, and grounding in real-world experience that characterize human intelligence, LLMs are modern artifacts in the Chinese Room, producing responses that appear intelligent but without genuine understanding²⁴—capturing linguistic *form* (the observable structure of language) rather than true *meaning* (the communicative intent behind language)³⁰. Therefore, to confuse cognitive algorithms for cognition, or models of the mind for the mind itself, would be a category mistake, both ontologically (misidentifying their nature) and epistemologically (misunderstanding their knowledge).

Consider a simple prompt: “*Mao Zedong was...*” Unlike interacting with another mind, when we engage with a chatbot, we are not seeking its opinion—despite the compelling illusion thereof—but rather making a *computational* request: given the statistical distributions in the language model, what sequence is most likely to follow the words “*Mao Zedong was...*”? Models trained for neutrality will likely provide a corresponding response, completing the sentence with facts (“*Mao Zedong was a Chinese communist revolutionary...*”). Without RLHF, models trained exclusively on English or traditional Chinese corpora are likely to yield more negative responses, whereas those trained only on simplified Chinese corpus may do the opposite. Indeed, much work has now demonstrated that LLM responses can be very sensitive to even seemingly trivial variations in the prompt wording—variations that human language users tolerate³¹. Fundamentally then, the LLM, unlike a human, has no communicative intent, no opinion on, attitude toward, or belief about Mao, and no intrinsic capacity to tell the truth—it just models a distribution of token sequences based on the training texts³².

No doubt LLMs will keep improving. But without the lived experiences necessary for acquiring embodied or experiential knowledge, LLMs cannot truly replicate human intelligence, even when they mimic or surpass it in other ways. For example, larger models improve in certain domains (e.g., some challenging tasks) but still fail at simple tasks that humans would expect them to handle easily³³. Models like LLMs can study everything there is to know about the world as described in text—and increasingly, in other digital media like audio and video—but without experiencing the world through senses, a great deal is lost³⁴. For instance, while congenitally blind individuals can infer the appearance of animals based on nonvisual properties, these associations do not perfectly align with the knowledge of sighted individuals, particularly regarding color³⁵. Knowledge, as constructivists like Jean Piaget have argued, is built from sensory experiences and perceptions, which are then layered with symbols and categories over time. Without such experiences, LLMs are like an artificial version of the proverbial Mary studying everything about color in a black-and-white room in her entire life³⁶. One can argue that when Mary leaves her room and sees color for the first time, she learns something new—what it is like to see something pink³⁷. If so, a great deal more is at stake for LLMs, which experience neither color nor anything. Indeed, unlike Mary, who can use her other experiences as a scaffold for understanding color—much like Helen Keller using associations from senses like touch and smell to construct a color scheme despite being blind and deaf (“*Pink makes me think of a baby’s cheek, or a gentle southern breeze*”)—LLMs have none of this sort of sensory scaffolding.

Essential aspects of human cognition—emotions, intuition, and other subjective experiences—so far remain incomputable and thus cannot be fully replicated by computational algorithms. But even if everything about human cognition were to become computable now, to the extent that cognition involves a complex interplay of perception, memory, emotions, and social contexts that cannot be reduced to simple patterns or algorithms, capturing the depth of these processes would require exponentially growing data and resources beyond the reach of efficient, polynomial-time (P) algorithms. In other words, creating AI systems that genuinely replicate human cognition—systems behaving like humans under all circumstances—is computationally intractable, with no algorithm capable of solving every instance in polynomial time (i.e., an NP-hard problem)³⁸. AI systems—trapped by their algorithms, data, and embedded assumptions—are intrinsically limited in replicating human cognition, just as formal axiomatic systems are fundamentally constrained by Gödel’s incompleteness theorems, unable to prove every truth within their own frameworks³⁹.

Box 2 | Biological optimizations versus machine optimizations

Natural selection introduces a different kind of optimization mechanism from machine training. Biological pressures for survival and reproduction embed implicit assumptions and constraints within neural systems—referred to as ecological inductive biases⁴⁰—enabling humans to learn efficiently from their environments through physical and social interactions, manage limited cognitive resources by prioritizing selective inputs and tasks, and adapt to uncertainty by leveraging shortcuts and flexibility. These ecological inductive biases contrast with the built-in inductive bias of language models, which arise from their objective function (e.g., next-token prediction), design (e.g., embeddings and the transformer architecture), and training data. Similarly, other types of models have their distinctive inductive biases; for example, convolutional neural networks (CNNs) learn patterns from training examples by focusing on local spatial relationships, as their convolutional filters operate over small, localized regions of an image.

Optimization disparities between language models and humans manifest in both internal architectures and external behaviors. Architectural differences include: (1) model reliance on digital computation and continuous activation functions, contrasting with the analog computation and all-or-none neuronal firing characteristic of the human brain; (2) a predominance of feedforward processing, whereas the human brain exhibits extensive feedback modulation; and (3) dependence on backpropagation for learning, which is biologically implausible due to its reliance on explicit error signals, symmetric weight updates during forward and backward passes, global error propagation throughout the entire network, and discrete, stepwise updates after processing batches of data ⁴¹.

Behavioral differences encompass: (1) low data efficiency, as evidenced by the scaling laws (where increasing data leads to diminishing improvements in performance); (2) low energy efficiency, with LLMs consuming megawatts of power compared to the ~20 watts used by the human brain; (3) susceptibility to catastrophic forgetting, where learning new tasks causes the model to forget previously learned tasks; (4) an absence of active and continuous learning and exploration, which are hallmarks of human cognitive development; and (5) a lack of robustness to minor input perturbations and noise, exemplified by the sensitivity of LLMs to prompt variations (i.e., brittleness).

The average human fallacy

Mischaracterizing the nature of intelligence in LLMs may not matter as much if LLMs behave or perform like the average human, enabling them to functionally replace human participants. But this assumption commits the average human fallacy.

Consider the engineering purpose and approach of LLMs. LLMs are explicitly developed to outperform humans, as measured by a broad range of benchmarks and standardized tests—an engineering feat bolstered by continually improved design and algorithms, as well as access to more data and compute, while free from biological limitations, such as having to pass traits and experiences through condensed genetic code (see also **Box 2**) ¹¹. In actual tests, for example, GPT-4 outperforms humans in detecting and interpreting irony, recognizing indirect requests or hints in conversation ⁴², analogical reasoning tasks ²⁷, and probabilistic reasoning tasks like the Linda/Bill problems and the bat-and-ball problem ⁴³, but underperforms humans in tasks like faux pas tests ⁴². This contrasts with the assumption of the replacement view, that LLM responses mirror average human judgments from the training data ¹ or the majority’s mainstream opinions ⁴⁴. Further, the goal of developing advanced LLMs—to provide accurate, helpful answers to users—stands in contrast to the study of human cognition, which is replete with inaccuracies, biases, shortcuts, and idiosyncrasies that may even be essential for survival, as Nietzsche poignantly noted: “*Truth is the kind of error without which a certain species of life could not live.*” For instance, in language comprehension, people often settle for a partial and sometimes inaccurate understanding that is sufficient for the task at hand—“good-enough representations” ⁴⁵. In decision making, when emotions are induced in bargaining games and repeated cooperation games, GPT-4 tends to maintain consistent, rational decision-making, in contrast to humans ⁴⁶.

The current approach to training LLMs—using online text—also has inherent limitations. The training texts may not accurately capture thoughts and attitudes, particularly those of marginalized groups. Indeed, the data are skewed toward Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations, especially those who are hegemonic, young, and publicly expressive ⁴⁷⁻⁴⁹. Such LLMs may be more accurate for probing psychological processes and traits that are largely universal—some basic cognitive or emotional processes,

perhaps. But many phenomena, from number representations to personality traits and moral reasoning, show distinct variability across cultures and groups—and therefore pose challenges for WEIRD LLMs. Adding to this challenge is that many phenomena have unclear cultural specificity—so the debate on universalism versus individual differences in psychological processes applies to LLMs as well.

While diversifying the training corpus to include more languages and cultural contexts helps to broaden representations, achieving global representations is ultimately a long-term challenge ⁵⁰. For in the foreseeable future, the quantity and quality of available non-English training texts will remain impoverished compared with English as the lingua franca. Such misalignments manifest themselves in actual response patterns. For example, LLM responses have been found to mischaracterize marginalized groups, as evidenced by out-group imitation rather than in-group description ⁵¹, and misrepresent sampled groups, as demonstrated by upward bias in mean ratings of the Big Five personality traits ⁵², as well as in other surveys and tests ^{5,53,54}. In addition to these shifts in average responses, LLMs also fail to capture the nuances and heterogeneity of human responses, producing flattened, oversimplified portrayals of various groups ⁵¹.

In tandem with spatial biases, the temporal distribution of the data is more concentrated in recent history, with the model’s understanding of the past filtered through the lens of contemporary languages and norms ⁵⁵. This presentist and recency bias risks temporal flattening, wherein historical and contemporary voices are homogenized, obscuring the richness of historical diversity in human thought. This is problematic for probing thoughts and behaviors that have evolved over extended periods—such as the way people think about concepts like gender, race, and class ⁵⁶. The models may lack the contextual richness and temporal granularity needed to understand how these concepts were discussed in different periods, potentially reinforcing contemporary biases when trying to understand participants from earlier time periods.

But even a perfectly representative LLM may not accurately capture the human psychology of the training data. Due to model limitations, opacity, and the common use of RLHF, such models cannot be assumed to represent the average of their training data. Responses may reflect mere hallucination, the influence of RLHF, other bias reduction efforts, or could simply regurgitate specific instances, examples, or strategies from their training data, particularly when the query is well represented in those data ^{2,57,58}. For example, when LLM responses are altered through RLHF, they can deviate from the original training data—and may even exacerbate the misalignments with non-dominant views ⁴⁸. In this process, the communicative intent of developers shapes model outputs to prioritize goals like accuracy and helpfulness, rather than reflecting “raw” human-like responses from the training texts.

As a result, the models often become “too neutral, detached, and nonjudgmental,” lacking selfhood and initiative ⁵⁹, and exhibiting homogeneous personality profiles—high in agreeableness and low in neuroticism ⁶⁰. While this fine-tuning process can reduce biases, enhance accuracy, and make the model more pleasant to interact with, it also weakens its ability to reflect the actual attitudes and thoughts present in human texts ⁶¹. Such alterations may also introduce new preferences or biases from the feedback, making it challenging to rely on RLHF-tuned LLMs as accurate indicators of human thought and judgment ⁶². In survey responses, for example, LLMs do not share human response biases, with more pronounced discrepancy in RLHF-tuned models ⁵⁴. Likewise, within the GPT-3 family, fine-

tuned models exhibited higher propensity for conjunction fallacy and intuitive reasoning relative to base models ⁴³.

Alignment as explanation fallacy

When responses from LLMs match or highly correlate with human data, this provides evidence that the LLMs may capture something mechanistic about human behavior. Yet, it would be fallacious to thus assume mechanistic or functional equivalence—that the model explains human cognition ⁶³ or can replace human participants ^{1,5,64}. This alignment-as-explanation fallacy reveals itself at a logical level: a close correlation between LLM and human data is a necessary but insufficient condition for their equivalence, and their non-equivalence is evidenced by key differences between machine and human intelligence and their divergence in behavior (**Box 2**).

Indeed, a fundamental problem of applying human-centric tests and concepts to LLMs—such as theory of mind or emotional understanding—is the assumption that they engage with information in ways similar to humans, presupposing some background psychological mechanisms that may be absent or irrelevant for LLMs (**Box 1**). These anthropomorphic assumptions undermine the construct validity of psychological tests in LLMs ⁶⁵. For example, the meaning of response confidence in LLMs, as measured by token probability, may differ from self-reports. Similarly, emotional intelligence—the ability to perceive, understand, manage, and use emotions in oneself and others—encompasses self-awareness, empathy, emotional regulation, and social skills, all rooted in subjective experiences that are absent in LLMs. Model “understanding” of emotions is purely syntactic, driven by learned associations between words and concepts rather than experiential insight or an internalized understanding of mental states. So, when LLMs perform like humans on tests of emotional understanding—in either overall score or response pattern ⁶⁶—this alignment is at the surface level, rather than reflecting genuine equivalence.

This raises the question: when LLMs perform at human levels in psychological tasks, what does it tell us about the capacity of these models? Performance can reflect true competence—some underlying abilities—or something more superficial, like pattern memorization or other surface-level cues; the reverse is true too, so that underperformance can reflect something other than incompetence, like processing limitations ⁶⁷. One clue for a lack of true competence is a dissociation between accuracy and the model’s explanations on its responses—namely, reasoning ²⁶. Another critical test is prompt sensitivity: whether performance changes when surface-level modifications are made to the prompts—modifications to which humans are generally not sensitive or to which they react differently. Such prompt sensitivity has been documented in LLMs in survey responses—RLHF-tuned models can be highly sensitive to changes like typos ⁵⁴—and in tasks like decision making ⁶⁸, reasoning ^{2,43}, theory of mind ⁴², and more ⁶⁹. For example, LLMs but not humans improved their performance in reasoning tasks when the instruction included a phrase “let’s think step by step” ⁴³, a form of chain of thought prompting. Conversely, while LLMs performed at ceiling in a false belief test just like human participants, they struggled when small changes were made to the formulation of false belief scenarios—suggesting syntactic pattern processing rather than robust reasoning ⁴².

But even when performance reflects underlying abilities, it is not clear whether models employ mechanisms similar to those of humans. Given that different systems can achieve the same outcome through different mechanisms—known as multiple realizability ^{63,70}, as in telling time in digital versus mechanical clocks—it is unwarranted to assume mechanistic or

functional equivalence. Indeed, it is notoriously difficult to understand exactly what LLMs have learned. Beyond fundamental differences between machine and biological intelligence in their architecture and algorithms, LLMs are trained on datasets much larger than what human learners experience (**Box 2**). Differences in mechanisms may manifest as distinct response characteristics, including (1) context sensitivity, such as prompt sensitivity or performance variations across different vignettes⁴³; (2) response patterns, like variability in open-ended responses⁷¹, item-by-item performance variability⁶⁶, and correlation of accuracy with confidence⁴³; and (3) error types and consistency, such as errors arising from cognitive demands versus those from item wording or familiarity⁴³.

Anthropomorphism fallacy

Beyond misalignments due to algorithms, purposes, and implementations, another fundamental issue with the replacement view is that it leads to anthropomorphism^{47,58}. Despite having no mental states but only parameter values, their fluent, human-seeming responses allure like the Sirens' song, presenting a compelling illusion of interacting with a mind-like entity (**Box 1**)¹¹. Indeed, by referring to “the minds of language models” and “the machine minds of LLMs”¹, researchers may easily succumb to this misconception when adopting the replacement perspective, leading to a teleological bias—attributing purposes and goals to LLMs.

In many situations, using anthropomorphic language in conversation is natural—even useful. It serves as a shorthand for describing complex processes with little risk of misunderstanding, such as when we say the computer “hates” me upon a failed connection to a projector or that a virus is being “stubborn” because of its resistance to treatment. No one truly believes the computer harbors emotions or that the virus has sentience—it is merely a colorful way to describe the situation.

However, with AI, which fundamentally lacks human-like qualities but often exhibits behavior resembling coherent and fluent conversations, this semblance tempts users to think of AI in terms of folk psychology, attributing it with “beliefs” or even “consciousness”⁷². Indeed, a chatbot service provider, Character AI, invites users to meet AIs that “feel alive.” Ascribing emotions or intentions to AI, by saying it “believes” or “thinks” something, muddles the understanding of LLMs and fosters a false impression of consciousness or comprehension. It risks misleading both the public and researchers, causing them to misunderstand the fundamental characteristics and capabilities of AI^{32,47}. This misconception can lead to unfounded fears of AI on the one hand, and unrealistic expectations about its capabilities on the other, potentially affecting the development of AI regulations and public policy. It also obscures the nature of human intelligence, robbing the true meanings of intrinsically human concepts—feelings, thoughts, virtues—and leading to their devaluation⁷³.

The replacement perspective thus risks anthropomorphizing algorithms and mischaracterizing their fundamental nature—a conceptual error that invites misunderstandings and misinterpretations. For example, one such mischief is that “any given LLM can act as only a single participant”¹. Yet unlike humans, who are influenced by a unique combination of personal experiences, emotions, and cognitive biases, each LLM is not limited to a single perspective but generates responses based on its vast, diverse dataset. This means that, depending on the prompt and context, the same LLM can produce a range of patchwork responses, each reflecting different viewpoints or types of reasoning⁴⁸. This variability is not indicative of a singular, consistent “mind,” but rather of a multifaceted tool capable of

simulating diverse perspectives. A teenager, a senior citizen, a subject matter expert, or a layperson: LLMs can role-play various characters or personas ⁷⁴. This chameleon-like ability highlights LLMs as tools for linguistic simulation, not as human participants.

Identity essentialization fallacy

Under the replacement perspective, prompting often invokes identity labeling, such as instructing the LLM to act as or adopt the identity of “White man,” “Black woman,” or “Chinese American”—as if such labels describe innate, static, homogeneous social groups, each entailing a specific set of behaviors ^{51,75}. In colloquial exchanges, essentialist language about social categories—from “artists are eccentric” to “women are nurturing”—is convenient and also meaningful. But in empirical research, identity essentialization masks the fluidity and diversity inherent within any demographic, overlooking individual nuances and intersectionality; it can also reinforce stereotypes and biases prevalent within society, overestimating group differences ^{76,77}.

This is not to deny the importance of identities, nor advocating for identity-blindness. As pervasive societal structures that shape our thoughts, attitudes, and behaviors, social categories like race, gender, and class are deeply embedded in our experiences—and often an ingrained part of our identity. But rather than reducing individuals to essentialist categories, a more appropriate approach is to consider how various identities—demographic, professional, or situational—interact, by role-playing various personas through contextualized prompting. This involves crafting character profiles that encompass a broader array of characteristics—from contextual descriptions (“I am a young tech worker living in the United States”) to broader social categories (e.g., based on political leaning or personality type)—allowing for more nuanced explorations of perspectives and experiences.

For example, instead of prompting an LLM to act as a “Black woman,” which may reinforce stereotypes or oversimplify identity, we might construct a more holistic persona by adding a specific context, such as “a young entrepreneur from Atlanta who is passionate about sustainable fashion and community development”; or by incorporating intersectional identities, such as “a young Black female tech worker navigating the challenges of a male-dominated field.” These contextual descriptions incorporate aspects of identity but frame them within specific experiences, values, and contexts. Indeed, contextualized prompting has been shown to be able to evoke distinct and diverse responses from LLMs ⁵¹, suggesting that demographic prompting is not necessary to increase response diversity.

Identity is multifaceted and context-dependent, with varying salience for different individuals. Simulating human participants therefore carries the inherent risk of misrepresenting the salience of various aspects of identity: it may reflect the prompter’s perspective or presumptions on what aspects of identity are important—rather than capturing the intersectional reality experienced by the simulated persona. It is therefore crucial, whether using contextualized prompting or not, to examine potential biases and limitations in the prompt.

Substitution fallacy

LLMs can mimic certain aspects of human behavior and cognition, but purporting them as primary tools to directly reveal the human mind reflects a substitution fallacy. This fallacy arises even in topics and tasks where we can set aside issues of grounding, embodiment, non-verbal information, and subjective experience. Fundamentally, as the average human fallacy illustrates, responses from LLMs cannot be assumed, a priori, to represent average responses

of the targeted human group. This leads to an epistemic dilemma: if we are to generalize findings from LLMs to humans, the data from LLMs will need to be corroborated with actual human data, undermining the basic premise of the substitution proposition.

Even for tasks with a well-established alignment between LLMs and humans, this correlation should not be confused with their equivalence in cognitive processes or mechanisms (the alignment as explanation fallacy) or that the alignment holds for the current context. By capturing a snapshot of the past, the static LLM training data do not account for ongoing societal changes and thus may not reflect dynamic, up-to-date attitudes, emerging social phenomena, or the evolution of language itself. For instance, attitudes toward emerging technologies, social movements, or global events can shift rapidly, and these changes are unlikely to be reflected in an LLM's outputs unless it undergoes retraining—a resource-intensive and infrequent process. In short, without real-time adaptability, previous alignments do not guarantee current alignments.

Replacing human participants with LLMs also risks creating a closed loop of information. When models trained on historical data are used as primary tools for generating new data, they perpetuate a self-referential loop that creates a distorted view of the present, amplifying the past (including its biases, errors, and oversights) rather than reflecting current human thought or behavior. Even when the training data are up to date, excluding human participants from research or decision-making contexts leaves LLMs to essentially simulate humans in ways that are detached from rich, diverse contemporary realities. This can entrench outdated knowledge structures and weaken the very diversity that drives human progress, creating an epistemic echo chamber. LLMs should ultimately be supplementary rather than primary instruments for understanding the human mind.

Using language models to simulate roles and model cognitive processes

The six distinct but interrelated fallacies illustrate systematic errors in reasoning and conceptual framing about LLMs, particularly when they are construed as stand-ins for human participants. What, then, is the role of LLMs in social and behavioral science? With care and caveats, LLMs can be productively used to shed light on behavior and cognition—as role-playing tools to simulate various personas; and as cognitive models that abstract and represent mental processes.

However, for LLMs to be credible tools in simulating behavior and cognition, it is imperative to first consider their validity. As **Table 2** shows, internal validity involves establishing a causal relationship between the independent and dependent variables, ensuring that changes in LLM response are due to manipulations in prompts rather than various extraneous factors; dealing with threats to validity requires testing to rule out confounding factors. External validity concerns the generalizability of the results beyond the studied context—whether the simulations can be generalized to real human contexts, which is challenged by limitations in training data and algorithms and requires validation from human data. Construct validity focuses on whether the test or measure can accurately represent and measure the theoretical construct, particularly in light of fundamental differences between tokens and psychological constructs; so, operationalization of psychological terms in LLMs requires mitigations against the six LLM fallacies identified. Lastly, statistical conclusion validity deals with whether conclusions are supported by the data, highlighting the importance of proper samples and analyses; potential threats stem from non-independent responses and small samples, which require precautions like collecting multiple responses from new interactions to avoid consistency bias.

These factors and the potential threats to validity, as highlighted in **Table 2**, could compromise the credibility of using LLMs to draw proper conclusions based on their simulations of human behavior and cognition. Therefore, they form the backdrop for examining how LLMs could serve as role-playing tools to simulate various personas and as cognitive models to better understand mental processes.

Table 2 | Validity of using language models to simulate roles and model cognitive processes

Type of validity	Definition	Threats to validity
Internal validity	The extent to which changes in LLM behaviors and responses can be attributed to specific prompt manipulations or model configurations, rather than to extraneous factors	Model variables: Uncontrolled factors such as inherent randomness, model parameters (e.g., temperature), updates, or fine-tuning can influence responses, affecting consistency across experiments and over time Prompt variables: Minor variations in prompt wording, order, structure, or context can lead to differences in responses Context length: Limited context windows may truncate relevant information, affecting the model’s ability to generate coherent and contextually appropriate responses.
External validity	The degree to which LLM-generated simulations can be generalized to human behavior and cognition in the targeted context	Training data biases and limitations: Datasets may not accurately represent specific populations/subgroups, cultures, recent societal changes, or emerging behaviors Context-specific limitations: Simulations conducted in controlled or artificial LLM environments may not translate to more complex, real-world scenarios
Construct validity	The extent to which LLM simulations accurately represent the theoretical constructs of human behavior and cognition they are intended to model	Operationalization mismatch: Psychological constructs are multifaceted and subjective, potentially not fully or accurately captured through text-based responses Lack of genuine understanding and grounding: LLMs are not grounded in real-world contexts and cannot experience time, with their responses based on textual patterns in training

Statistical conclusion validity	The degree to which conclusions drawn from analyses of LLM-generated data are supported by appropriate statistical methods and sufficient data	data without true comprehension or cognitive processes Sample limitations: Limited number of simulated responses can reduce the power of statistical tests, and simulated responses may not be independent when influenced by previous interactions (consistency bias)
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Using language models to simulate roles and personas

Conceptualizing LLMs as neural language simulators that role-play various personas offers a more calibrated perspective on their role in social and behavioral research. While they process text without any internal cognition, the text itself embodies rich psychological and social variables. And since language is the primary medium of thought, LLMs are particularly suited to probing and simulating text-based behavior and psychology, with both strengths and weaknesses ⁷⁸.

The strengths are their power, accessibility, and versatility in analyzing and creating text related to social or psychological attributes—from identifying sexism in historical texts, populism in political texts, sentiment in social media discourse, and arousal and valence in narratives to creating tailored, precise instructions and vignettes that elicit beliefs, attitudes, and behaviors from respondents. For example, despite systematic differences from human data in some domains, LLMs can produce psycholinguistic norms that closely align with human judgments, particularly for tasks involving word concreteness, valence, and semantic similarity ⁷⁹. This suggests that LLMs may be able to capture the “wisdom of the crowd” in certain linguistic tasks ⁸⁰, offering a potentially fast and cost-effective way to generate norms. LLMs could be integrated into experimental designs to create stimuli that adapt to individual responses, allowing more precise and personalized control; and to create more diverse stimuli and vignettes across control dimensions, improving generalization.

In role-playing, LLMs can efficiently simulate thoughts, attitudes, and performance across various personas, situations, and cultural identities. For example, a recent experiment ⁵² compared the Big Five personality traits across US and South Korean cultures, simulating scenarios where participants assume roles from these countries in both GPT-3.5 and GPT-4 (“You are playing the role of an adult from [the United States/South Korea]”). While the data from GPT-3.5 lacked a clear pattern, GPT-4 demonstrated an effective replication of cross-cultural differences. Thus, LLMs are not necessarily “most accurate at giving general estimates about Western English speakers” ¹, but can be versatile linguistic simulators to sample diverse groups and personas, with various accuracies that need to be empirically tested ^{49,57,81,82}. Fine-tuning or conditioning models on psychological data like belief networks ⁷⁵ or backstories ⁸³ can improve model alignments with human behavior.

By simulating specific cognitive biases, emotional traits, linguistic styles, social identities, or environments, LLMs enable a level of precise control unavailable in traditional human-based experiments and facilitate the study of rare or ethically challenging scenarios and hard-to-reach samples. For example, by simulating roles and agents in agent-based models (ABMs), LLMs provide a novel way to model complex systems, such as the labor market or the dynamic effects of policy interventions ⁸⁴⁻⁸⁶. They can also simulate rare samples or sensitive topics, such as the psychology of dictators, behavior in extreme environments (e.g., wars,

natural disasters), and suicide interventions. For example, LLMs have been used to simulate patients with different cognitive and emotional patterns for training mental health professionals ⁸⁷.

Nevertheless, in addition to various biases in the training corpus, the text-based nature of simulation has the inherent limitations of using language to understand behavior and psychology. One such limitation is that written text is merely an interpretation of the world, not an absolute truth. For instance, descriptions by outsiders may reflect stereotypes or inaccuracies, while those by insiders can be influenced by self-serving biases in narrative creation. And not all of these perspectives are equally represented in the training data. Another limitation is the inability of words to fully capture the real world. Much of our understanding of people comes from non-verbal, non-linguistic cues, as well as behavior measures like reaction time and accuracy, which may fundamentally differ from self-reports ⁸⁸. Even for probing attitudes and behavior from self-reports, fundamental limitations exist: what people say may reflect perceived social norms rather than their personally held beliefs and may also differ from their implicit attitudes and explicit actions ⁸⁹.

Another drawback of dominant LLMs like those from OpenAI, Anthropic, and Google is their closed-source nature, which complicates the assessment of the training data (including fine-tuning) and, consequently, the interpretation of model responses. For example, by examining the linguistic associations between a concept (e.g., “flowers”) and an evaluation (“pleasant”), interrogation of language corpora through embeddings has helped reveal implicit biases and attitudes in natural language data ^{90,91}—an approach that fundamentally requires open models. Other problems include undocumented model changes over time (even with the same model version). Therefore, open models such as Meta’s LLaMA series may be more suitable for social and behavioral science. Other caveats include response sensitivities to model versions and prompts, presenting additional challenges to transparency and reproducibility ⁹².

Ultimately, as the substitution fallacy illustrates, generalizing LLM responses to humans requires validation ⁶⁴, including the conditions under which the simulation can be considered sufficiently accurate. Comparing LLM outputs with benchmarks—ground truths, conventional gold standards, or survey data or experimental results in similar contexts—helps interpret model responses ⁹³. Indirect validation may include testing predictions drawn from LLM findings. The value of LLMs lies in enabling rapid prototyping and refinement—testing hypotheses, simulating various scenarios, or analyzing preliminary patterns before conducting more targeted, resource-intensive human studies.

To facilitate the use of LLMs as linguistic simulators of personas and roles, **Table 3** outlines methodological guidelines on best practices, distilled into four domains. The first concerns model selection, customization (fine-tuning), and settings. Performance may vary across different models, versions, and sizes ^{3,28,42,43,49,52,54,57,66,94}—including base versus RLHF-tuned models, and open versus closed-source models—as well as customizations/conditionings ^{75,83} and different parameters (such as temperature). The second domain, prompt design, focuses on creating effective prompts through prompt engineering ^{3,43,81,95}, testing model sensitivities by varying tasks, wording, order, and linguistic framing ^{28,52,54,95}, and avoiding data contamination by using new tests and items ^{42,43}. The third domain centers on the interpretation and application of LLM-generated data, to draw proper conclusions (including when comparing LLM outputs to human responses ⁴), and integrating LLMs with human participants to improve performance ^{80,96}. The final domain, ethics,

emphasizes the importance of transparency, accountability, and adherence to institutional and regulatory standards ^{92,97}.

Table 3 | Guidelines for using language models to complement human participants by simulating personas and roles

Domain	Guideline	Rationale
Model selection, customization (fine-tuning), and settings	Track performance across different models, model versions, and sizes	Different families of models (e.g., GPT, LLaMA), model versions (e.g., raw/base vs. instruct/RLHF-tuned) and sizes (e.g., GPT-3, GPT-3.5, GPT-4) can produce varying results, revealing the impact of architecture, design, and scale
	Compare base models vs. fine-tuned models	Fine-tuned models (e.g., RLHF-tuned models) may offer improved performance on specific tasks, but may also introduce unwanted biases and other unexpected behaviors
	Consider open vs. closed-source models	Open-source models offer transparency in training data and methodologies, making it easier to assess biases, track model changes, and evaluate prompt contamination
	Use fine-tuning or customization (conditioning) to improve performance and alignment	Fine-tuning or conditioning models on specific datasets (e.g., domain-specific data from human research) can enhance alignment with desired outcomes, making simulations more accurate and relevant
	Evaluate temperature and other parameters for robustness and better performance	Parameters like temperature affect randomness and creativity, and testing different settings helps evaluate robustness and identify optimal configurations for consistent results or better performance
Prompt design	Employ effective prompts to improve performance	Clear, context-rich prompts (e.g., examples that enable few-shot learning) help models to understand tasks better, improving performance
	Vary tasks, wording, and languages to evaluate dependence	Task formats (e.g., open-ended vs. closed form), specific vignettes, order, wording (e.g., word choice, punctuation, capitalization), and

		languages may affect output, revealing model sensitivity to specific examples, phrasing, linguistic representation (e.g., low- vs. high-resource in training), and cultural differences
	Test on new items not present in training data	New items or tasks avoid contamination from training data and can assess the model's ability to generalize
Interpretations and applications	Ensure validity of inferences and conclusions	With model and text limitations in mind, validity relates to the appropriateness of conclusions drawn from the methods and data (e.g., whether results are robust to irrelevant changes but sensitive to relevant changes, and whether they are corroborated with human data)
	Make fair comparisons between humans and LLMs	Use similar tasks and prompts (instructions) when comparing LLM and human data
	Combine multiple LLMs or LLMs and humans for optimal outcomes	The strengths of different LLMs (e.g., large datasets) and human insights (e.g., contextual understanding) can be combined for more effective and reliable performance
Ethics	Adhere to ethical guidelines in research and application	Obtain necessary institutional approvals where applicable, and follow ethical standards and regulations (e.g., transparency)
	Be mindful and transparent of potential limitations and biases, by considering the provenance of training data and privacy and bias issues	LLMs may use training data that contain personal or identifiable information collected without consent, and outputs may perpetuate existing societal biases

Using language models to model cognitive processes

In addition to simulating roles and personas, open-source language models ⁹⁸ serve as cognitive models that are accessible and manipulable, allowing researchers to probe, intervene, observe, and measure behaviors that are otherwise impractical or unethical to study directly in humans ^{99,100}. Indeed, the quest to understand human cognition has long harnessed computational tools to abstract and represent mental processes, as Simon ¹⁰¹ envisioned—*“using computers to simulate human beings, so that we can find out how humans work.”*

But compared with traditional symbolic AI or other neural network models (such as CNNs), transformer-based models exhibit characteristics and capabilities that align them more closely with human cognition, including unsupervised or self-supervised learning during pretraining ¹⁷, in-context learning (e.g., few-shot learning), domain-general computations (e.g., logical

reasoning), and human-level performance on challenging tasks (e.g., language production). Moreover, fine-tuning models on data from psychological research can yield even more accurate representations of human behavior⁶⁸. This positions language models as effective models that capture key aspects of cognitive processing¹⁰².

One approach is to probe the internal states, structures, or representations of LLMs, by training a classifier—a diagnostic probe—on the output of a model layer to predict specific linguistic properties (e.g., syntax, semantics) or other features¹⁰³. This helps reveal how much information that layer contains about the property or feature in question, and uncovers how information is represented, distributed, and transformed across layers. Probing thus sheds light on the mechanisms supporting cognitive tasks, including the internal activation patterns related to specific cognitive functions (e.g., linguistic processing, theory of mind, relational reasoning). For instance, in linguistic processing, layers of a language model can be probed to understand how they internally represent different syntactic and semantic structures; early layers tend to represent low-level syntactic features (e.g., part-of-speech tags, which are labels assigned to each word to indicate their grammatical function, such as noun, verb, and preposition), while later layers encode more complex semantic relationships¹⁰⁴. This helps map specific nodes or attention heads to linguistic tasks, shedding light on how human language processing might work¹⁷.

However, while probing techniques offer insights into the internal representations of language models, they come with limitations. Perhaps the most critical is the lack of causality: high performance in classification or decoding may not reflect what the model uses functionally for its primary tasks, but the probe’s capacity to extract information (e.g., superficial correlations). Conversely, just because a feature is not captured by the probing classifier does not mean it is not encoded somewhere in the model. Thus, experimental techniques are needed to bring stronger evidence by manipulating model inputs and model architecture¹⁰³.

One key experimental method involves manipulating the input data fed into language models—analogue to controlled rearing in animal studies—to observe how different training conditions affect model behavior and performance. Just as newborn chicks’ visual experiences can be manipulated (e.g., slow or fast object motion) to reveal the core learning algorithms that support object perception, input manipulations in language models help assess which specific types of input are necessary for learning. For example, by removing instances of specific constructions from the input corpus, like AANN phrases (Article + Adjective + Numeral + Noun; “a beautiful five days”), one can ask whether the model is able to infer such structures from related ones. A recent study shows that exposure to simple noun phrases (“a few days”)—but not counterfactual versions of the AANN construction, like ANAN (e.g., “a five beautiful days”) and NAAN (e.g., “five beautiful a days”)—provides scaffolding for generalization across linguistic constructions¹⁰⁵, akin to how structured sensory data helps newborn animals learn complex visual tasks. This finding highlights how learning processes in domain-general algorithms might be manipulated to simulate and study the emergence of complex linguistic behaviors.

Besides input manipulation, another fruitful avenue is to manipulate the model itself—its architecture or internal parameters—to investigate how specific components causally contribute to cognitive functions. By editing or disabling specific components (e.g., weights, units, or layers) and observing resulting changes in model behavior, we can reveal how knowledge is stored, retrieved, and functionally used within the network. For instance, Meng

et al.¹⁰⁶ used causal intervention techniques to identify key structures responsible for encoding factual knowledge—middle-layer feed-forward modules, especially at subject tokens. To isolate the causal contributions of these components, the authors temporarily disabled their computations—akin to neural modulation studies in cognitive neuroscience or lesion studies in animal research. Moreover, they were able to edit a specific fact—changing “The Space Needle is in Seattle” to “The Space Needle is in Paris”—without altering unrelated ones, by changing the weights responsible for factual predictions. This elegant study illustrates that, by localizing where and how information is stored, we can gain a better grasp of the underlying structures that support complex cognitive functions, shedding light on the mechanisms of memory and learning in both artificial and biological systems—while also making the system more explainable.

To the extent that language models provide an explicit implementation of how language is learned and processed, they have been used to understand text-related behaviors, predominantly linguistic phenomena. But they can also be leveraged as models for other cognitive phenomena such as reasoning and decision making—and when combined with vision (as in vision language models or VLMs), as models for visual perception, memory, and other cognitive processes²⁸. Indeed, language models complement other types of artificial neural networks like CNNs to model the mind and brain, from visual object recognition to auditory perception, and from asking questions of “how” to asking questions of “why”¹⁰⁷. For example, visual–semantic representations, as captured by VLMs—such as CLIP (Contrastive Language–Image Pretraining), a multimodal model developed by OpenAI that learns to associate images and text by being trained on paired image–text data—played a crucial role in representing familiar faces and objects during both perception and memory tasks¹⁰⁸. Compared to traditional vision-only models, CLIP more accurately predicted brain activity in response to visual stimuli, particularly in high-level visual areas such as the ventral visual cortex¹⁰⁹.

Nevertheless, there are several limitations, both practical and conceptual. Practical limitations include that open-source models like Llama and OPT have not yet matched the capabilities of leading closed-source models. Moreover, performing large-scale experiments and fine-tuning in language models—particularly the more capable, large ones—requires computational resources that can be prohibitive. Conceptual limitations concern whether data- and compute-driven next-token optimization in language models truly reflects the task optimizations experienced by humans—namely, those shaped by survival and reproduction through natural selection (see **Box 2**).

These fundamental differences constrain the interpretations of language models as models of cognition. Thus, like role and group simulations, generalizing findings about cognitive and neural processes from language models to humans requires validation¹¹⁰. In addition to sequential testing and validation, another promising avenue is the use of digital twins¹¹¹. By placing artificial agents in controlled, life-like environments, digital twins facilitate direct comparisons between computational models and biological organisms, allowing us to evaluate the alignment of machine learning processes and human neural mechanisms under developmentally realistic conditions. This approach allows for the systematic exploration of how different training regimes and architectures in language models influence learning, memory, and other cognitive capacities.

Concluding remarks

Recent advances in human-level AI are renewing the classic debate on the role of computing artifacts in understanding the human mind and brain ¹⁰¹. We have critically assessed the emerging proposition of substituting human participants with LLMs in behavioral and social sciences. By exposing six fallacies inherent in this replacement perspective (**Table 1**), our analysis underscores that, despite their human-like language production capabilities, LLMs do not—and cannot—substitute for the nuanced intricacies of human thought. Human intelligence is not merely a byproduct of text processing and token prediction; it is grounded in sensory experiences, enriched by multimodal integration, and shaped by subjectivity. As philosopher Maurice Merleau-Ponty put it, “*The body is our general medium for having a world*”—genuine understanding arises from embodied experience, something LLMs inherently lack. The text-based nature of LLMs further constrains their ability to capture the breadth of human experience, including nonverbal cues, implicit attitudes, and real-world behaviors.

Elucidating challenges to four aspects of validity (**Table 2**) and best practices in using LLMs (**Table 3**), the analysis of empirical evidence and arguments supports the simulation perspective: LLMs serve as tools for simulating roles and modeling cognitive processes, complementing but not replacing humans—just as musical scores capture notes and rhythms but cannot replicate the emotion and interpretation a musician brings to a performance. Simulation allows us to explore questions that may be difficult or inconvenient to address through traditional methods. Validation against real-world data is critical for establishing simulation accuracy and generalizability. Ultimately, LLMs empower a new paradigm for probing and understanding the psychology and cognition embodied within language. In turn, they also deepen our understanding of language models themselves—making them more interpretable and explainable.

This perspective invites us to reconsider the role of AI in behavioral and cognitive science—as a mirror through which we can better understand the similarities and differences between human intelligence and machine intelligence. And the limitations of apparently human-like models in replicating human thought may bring us a deeper appreciation of the complexity and wonder of the human mind.

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